

In-class work + Mini mock

Human Reproduction · Year 9 Science

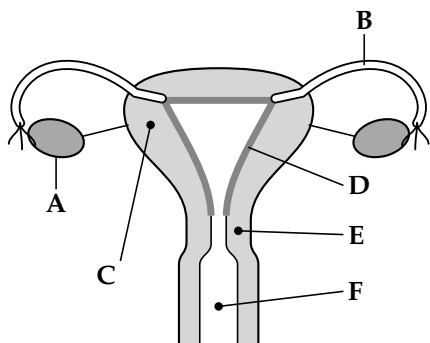
Each question: A / M / E

Part A after Loop 1, Part B after Loop 2, Part C after Loop 3, Part D after Loop 4. Part E is the mini mock at the end of the lesson.

Part A · Sex cells and the two systems

1 The diagram shows the female reproductive system. Write the letter of the part for each sentence.

[A]



(a) Eggs are made here.

(b) Fertilisation happens here.

(c) An embryo implants in this.

(d) Sperm are received here.

(e) This is a ring of muscle.

2 Complete the sentences about the male reproductive system.

[A]

Sperm are made in the _____, which are held outside the body in the _____. The sperm are stored in the _____. They travel along the sperm _____, where glands add fluid to make _____. The semen leaves the body through the _____.

3 The middle piece of a sperm cell is packed with mitochondria. Explain how this helps the sperm cell to do its job.

[M]

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4 An egg cell is one of the largest cells in the body. A sperm cell is one of the smallest. Explain why each size suits the job of the cell.

[E]

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Part B · Hormones and the menstrual cycle

5 Name the four stages of the menstrual cycle, in order, starting with the period.

[A]

1. _____ 2. _____ 3. _____ 4. _____

6 Write the name of the hormone that does each job. **[A]**

(a) It makes an egg mature in the ovary.

(b) It triggers ovulation.

(c) It builds up the lining of the uterus after a period.

(d) It keeps the lining of the uterus thick after ovulation.

7 The table shows the level of progesterone in a woman's blood during one cycle.

Day of the cycle	2	6	10	14	18	22	26	28
Progesterone level (units)	1	1	1	2	9	14	6	2

(a) Describe how the level of progesterone changes during the cycle. **[M]**

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(b) Explain what the fall after day 22 tells you about this cycle. **[E]**

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Part C · Fertilisation, chromosomes and twins

8 Write these in the order in which they form, earliest first: **embryo, foetus, sex cells, zygote**. **[A]**

_____ → _____ → _____ → _____

Number of chromosomes in a human sperm cell: _____ in a human zygote: _____

9 A sperm cell carrying a Y chromosome fertilises an egg cell. **[A / M]**

(a) Write the two sex chromosomes of the zygote.

(b) State the sex of the baby.

(c) Explain why it is the sperm cell, not the egg cell, that decides the sex of a baby.

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10 Ana and Bea are twin sisters, but they do not look alike. Explain how these twins formed and why they are not identical. **[E]**

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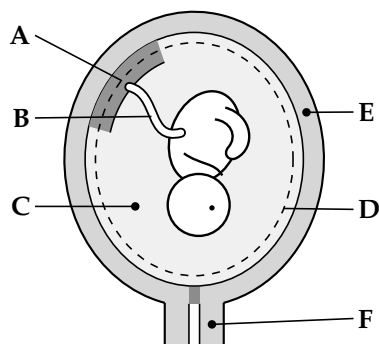
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Part D · Pregnancy and birth

11 The diagram shows a foetus in the uterus. Write the letter of the part for each sentence.

[A]



(a) It cushions the foetus.

(b) It joins the foetus to the placenta.

(c) Substances pass to and from the mother's blood here.

(d) It opens (dilates) during labour.

12 Urea is a waste product made by the cells of the foetus. Explain how urea leaves the blood of the foetus.

[M]

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13 The table shows the mass of a foetus during pregnancy.

Week of pregnancy	8	16	24	32	40
Mass of foetus (g)	1	120	640	1760	3400

(a) Calculate the increase in mass between week 32 and week 40.

[A]

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(b) Describe how the mass of the foetus changes during pregnancy.

[M]

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(c) A baby born at week 32 usually needs special care in hospital. Use the table to suggest why.

[E]

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Part E · Mini mock (12 minutes)

14 State what is meant by **fertilisation**.

[A]

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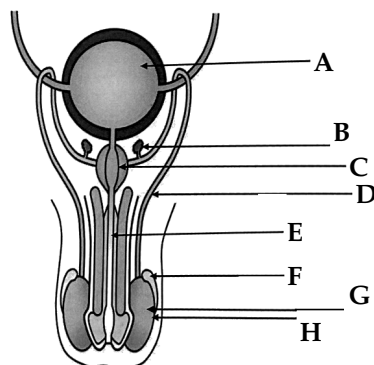
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15 State the number of chromosomes in:

[A]

(a) a human egg cell _____ (b) a human skin cell _____

16 The diagram shows the male reproductive system. Write the letter of the part for each sentence. [A]



(a) Sperm are made here.

(b) Sperm are stored here.

(c) This tube carries urine and semen.

(d) This keeps the testes cool.

17 Explain why the lining of the uterus becomes thick every month.

[M]

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18 A student is asked to describe implantation.

Sam's answer

Implantation is when the sperm joins the egg in the uterus. The ball of cells then sinks into the wall of the oviduct.

Sam's answer has 2 errors. Find each one and write the correct science.

[M]

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19 A pregnant woman is advised not to drink alcohol. Explain how alcohol in her blood can reach the foetus, and why it is harmful. **[E]**

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20 The table shows how often IVF treatment led to the birth of a baby, for women of different ages.

Age of woman (years)	under 35	35–37	38–39	40–42	43–44
Treatments that led to a birth (%)	32	25	19	11	5

(a) Describe the pattern in the table.

[M]

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(b) Suggest one reason for this pattern.

[E]

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